

1).

Constraint Satisfaction problem

Given

Doctors : D_1, D_2, D_3

Shift:

* Morning (M)

* Afternoon (A)

* Night (N)

Constraints

* D_1 cannot work Night* D_2 must work before D_3 (Morning < Afternoon < Night). —

* Only one doctor per shift.

* D_3 cannot ~~work~~ work Morning

* Every doctor gets exactly one shift

Step 1: Assign $D_1 \rightarrow$ Morning

Remaining : Afternoon, Night

Step 2: Assign $D_2 \rightarrow$ Afternoon

Remaining : Night

Step 3: Assign $D_3 \rightarrow$ Night

Check Constraints

* D_1 not Night ✓* D_2 before D_3 ✓* D_3 not morning ✓

* One doctor per shift ✓

* Every doctor assigned One ✓

2) Robot Grid Search (BFS)

Constraints

- * Move: Up, Down, Left, Right
- * Cost = 1 per move.
- * Cannot cross obstacles
- * Manhattan Distance heuristic.

Assume Grid:

```

S . . . X . .
. X . X . .
. X . . .
. . X X .
X . . G .
    
```

BFS Node

Queue: [S]

Expand S:

Queue = [(1,0), (0,1)]

Expand (1,0)

Queue = [(0,1), (2,0)]

Expand (0,1)

Queue = [(2,0), (0,2)]

Expand remaining node until Goal reached

Optimal path

```

S
↓
↓
→
→
↓
↓
→
G
    
```

Path Cost

Number of moves = 7

Total Cost = 7

3).

a). Effect of Constraints:

- * Limited Movement: Robot moves only in four direction
- * Obstacles: Robot must avoid blocked cell.
- * Limited Sensing: Robot knows only nearby cell.
- * Movement Cost = 1: Every move increase total Cost.
- * Risky zones Cost = 2: Robot prefers safer path.
- * Dynamic Environment: Robot updates knowledge while moving.

These Constraints help help the robot to choose the safest and cheapest path.

b). Solution Approach:

Steps:

1. Starts from S.
2. Observe Adjacent Cell.
3. Store safe cell in the frontier.
4. Expand the node the lowest total Cost.
5. Ignore blocked Cell.
6. Add extra cost for risky zones
7. Update the map whenever new information is discovered.
8. Continue until reaching G.

c). AI Concepts

Intelligent Agent:

- * perceives surrounding using sensors.
- * Makes decision automatically.
- * Acts to reach the goal.

Rationality:

- * Chooses action that minimize total cost.
- * Avoids obstacles and risky areas.
- * Selects the safest & shortest route.

Environment Type:

- * partially observable
- * Dynamic
- * Sequential
- * Discrete
- * Single-Agent

d). Justification:

- > It always finds the lowest-cost path
- > It handles different movement costs.
- > It guarantees an optimal solution under the given constraints.